Assignment: Variational Autoencoders for Remote Sensing

Foundational Models and Generative AI

Dataset: EuroSAT (64x64 RGB Satellite Images)

1. Objective

The goal of this assignment is to build a generative AI model capable of synthesizing satellite imagery from scratch. You will develop a Variational Autoencoder (VAE) to learn the complex textures of the Earth's surface (such as forests, rivers, and industrial zones). You must implement the neural architecture and the probabilistic loss function and perform advanced mathematical operations in the model's "hidden memory" (latent space) to demonstrate that the model has learned semantic concepts.

2. Implementation Tasks

Task 1: Architectural Design [1 mark]

You are required to build a complete VAE architecture using PyTorch.

The Encoder: Design a Convolutional Neural Network that takes a 64x64 RGB image and compresses it into a latent vector of size 32. It must output two distinct vectors: the Mean and the Log-Variance of the latent distribution.

- **The Bridge:** Implement the "Reparameterization Trick" function. This mathematical step is crucial: it allows the model to sample random noise from a Normal distribution using the learned Mean and Variance while ensuring the network remains differentiable for backpropagation.
- **The Decoder:** Design a Transposed Convolutional Network that takes the sampled latent vector and upsamples it back into a full 64x64 RGB image.

Task 2: Training and Loss Function [1 mark]

Implement the specific VAE objective function and train your model on the EuroSAT dataset.

- **The Loss Function:** You must write a custom loss function that combines two competing terms:
    1. **Reconstruction Loss:** Measures how accurately the output image matches the input (using Mean Squared Error).
    2. **KL Divergence:** Measures how closely the learned latent distribution matches a standard Normal distribution. **Note**: *Ensure you derive or implement the formula correctly, specifically paying attention to the sign (penalty vs. reward).*

- **Training:** Train the model for at least 15 epochs. The generated images must show recognizable textures (e.g., the grid patterns of highways or the green texture of forests) rather than just gray noise.

Task 3: Latent Space Interpolation **[1.5 marks]**

Test the model's ability to "morph" between concepts by navigating its hidden memory.

- **The Procedure:** Select two distinct images from the test set (e.g., an "Industrial Plant" and a "Pasture"). Encode them to get their latent vectors. Generate a sequence of 10 decoded images by calculating the linear intermediate steps between the first vector and the second vector.
- **Deliverable:** A single plot displaying this 10-step transition, showing how the structural features of the factory fade or transform into the open field.

Task 4: [Advanced] Semantic Vector Arithmetic **[1.5 mark]**

Demonstrate that your model has disentangled semantic features by performing algebra on the latent vectors.

- **The Experiment:**
    1. Calculate the average latent vector for a batch of "Forest" images.
    2. Calculate the average latent vector for a batch of "Highway" images.
    3. Subtract the Forest vector from the Highway vector. This result is your "Paving Direction" vector.
    4. Take a new image of a "River," encode it, add this "Paving Direction" vector to its latent code, and decode the result.
- **Analysis:** Observe the result. Does the river texture change to look like a road or canal? Does the shape of the river remain?

3. Submission Requirements

Please submit a single Colab Notebook containing:

Submit the <Roll No.>.ipynb file (do not add a Colab link) with proper visualization and outputs.

1. **Full Source Code:** Your implementation of the Encoder, Decoder, Reparameterization Trick, and Loss Function.
2. **Training Curves:** A graph showing the Loss decreasing over the epochs.
3. **Generated Samples:** A grid of 16 images generated from random noise.
4. **Interpolation Plot:** The visualization from Task 3.

Executive Summary: A brief report (max 300 words) analyzing the result of Task 4. Discuss whether the model successfully applied the "Paving" texture to the "River" and what this implies about the model's understanding of land use features.

This should be submitted in a one-page report named as <Roll No.>.pdf.

4. Submission Guidelines:

Due Date: 22/02/2026 till **11:59 P.M.**

Penalty for late submission: 1 mark deducted per day after due date.

Any deviation from the above instructions in submission will result in a heavy penalty. Once submitted, no changes or additions/deletions will be allowed.

Generic, text-only explanations without supporting plots, images, or experimental results will not be evaluated favorably. All figures included in the report must be generated using your own code.

Plagiarism: DO NOT plagiarize from the internet or your peers. The institute's plagiarism policy will be strictly enforced.

AI Usage: Do not use LLMs to solve this assignment. We value your logical derivation and coding ability over generated outputs.

Proper comments should be given

Note: No need to comment after submitting the assignment; we will be notified automatically after you turn in the assignment.